[INCH-POUND]
AA-P-2930
July 15, 1997
SUPERSEDING
MIL-P-28636B
15 February 1985

FEDERAL SPECIFICATION

PANEL FOR BUNK BED

The General Services Administration has authorized the use of this specification by all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers the requirements for laminated surfaced particleboard bunk bed headboard and footboard panels. This specification covers one classification of panel.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard

FED-STD-595 - Colors Used in Government Procurement.

(Copies of federal specifications and standards required by contractors in connection with specific procurement functions are obtained from the Defense Automated Printing Services, Attn: DoDSSP, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commanding Officer (Code 15E2), Naval Construction Battalion Center, 1000 23rd Avenue, Port Hueneme, CA 93043-4301, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7105

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI - A208.1 - Wood Particleboard.
ANSI - Z1.4 - Procedures, Sampling and Tables for Inspection by Attributes.

(Application for copies should be addressed to the American National Standards Institute, Inc., 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 4317 - Standard Specification for Polyvinyl Acetate-Based Emulsion Adhesives.
ASTM D 4690 - Standard Specification for Urea-Formaldehyde Resin Adhesives.

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA LD 3 - High-Pressure Decorative Laminates.

(Application for copies should be addressed to the National Electrical Manufactures Association, 2101 L Street N.W., Washington, DC 20037.)

3. REQUIREMENTS

3.1 Description. The bunk bed panel shall consist of a particleboard core surfaced on each side with a decorative laminated sheet. Two panels are required for each bunk bed, one to be used as a headboard and one as a footboard.

3.2 First article. When specified (see 6.2), a sample bunk bed panel shall be subjected to first article inspection (see 4.2.1).

3.3 Standard commercial product. The bunk bed panels shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the bed panels being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.4.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and finished product, provided form, fit, and function requirements are satisfied.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.5.1 Particleboard. The panel core shall be particleboard in accordance with ANSI A208.1, Grade M-3, weight of 45 pounds per cubic foot (720.9 kilograms per cubic metre), and a minimum of 0.625-inch (15.88 millimetre (mm)) thick. The particle board shall be laminated and have edge molding to decorate and protect the particle board from damage. When solid color laminated sheets are used, the edge molding color shall be of the same color or a lighter shade of the same color.

3.5.2 Decorative laminated sheets. Decorative laminated sheet used for surfacing bed panels shall be in accordance with GP 32 of NEMA LD 3. Pattern, color, and finish shall be as specified (see 6.2).

3.5.3 Adhesive. Adhesive for bonding high pressure decorative laminated plastic sheets to panel cores shall be in accordance with ASTM D 4317 or ASTM D 4690.

3.5.4 Clamps. Clamps and all necessary hardware shall be provided for securing the panels to the 0.875-inch (22.23 mm) bed cross tubes, as shown on figure 1. The clamp shall be a full-loop closed clip, but furnished open for slipping over the cross tubes. Unless otherwise specified (see 6.2), the PVC coating color shall conform to color no. 16251 (gray) of FED-STD-595.

3.5.5 Label holder. When specified (see 6.2), a label holder capable of holding a card or label with a tolerance of 2.75 inches wide by 1.5 inches high (69.85 mm by 38.10 mm) shall be provided. One label holder shall be provided for each set of two bed panels.

3.6 General construction. The design and dimensions of the bed headboard and footboard panels shall be as shown on figures 1 and 2 and as specified hereinafter. Unless otherwise noted, dimensions are minimum. Materials shall be in accordance with the details specified in 3.5 and hereinafter.

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3.6.1 Panel construction. The panels shall be constructed of particleboard covered on each side with one piece of decorative laminated sheet. Adhesive for securing laminated sheets to particleboard shall be in accordance with ASTM D 4317 or ASTM D 4690 (see 3.5.3). When label holders are specified, fasten a label holder with not less than three escutcheon pins to one of each set of two bunk bed panels at the location shown on figure 2. Finished laminated sheets shall not be cracked, split, scratched, scuffed, peeled, dented, or contain ripples. The entire edge of each panel shall be covered. The molding shall cover the panel edge including the laminated sheets, or may cover the edge to within 0.0156-inch (0.396 mm). Each panel shall have but one edge molding joint at the center of the bottom. Ends of the edge molding shall be square cut, meet flush, and be secured by brads at each side of the joint. The brads shall be driven from one side panel, set and filled to match the color of the laminated sheet. The edging shall fit tightly and securely, without gaps or crevices. Finished edge molding shall not be cracked, split, scratched, scuffed, or dented affecting serviceability or appearance.

3.6.2 Instruction sheets. Unless the manufacturer of the bed panels is required to install the panels on the bunk beds (see 6.2), two printed instruction sheets for installation of panels on the bunk beds shall be furnished in each carton of panels and in each bulk package of hardware (see 6.7). The sheet shall provide all necessary installation information and shall include a sketch of the completed unit and a cross-section view at the clamp similar to that shown on figure 1.

3.7 Workmanship. Workmanship shall be the highest quality throughout in accordance with the best standard practice used in the manufacturer of furniture of this type. There shall be no open joints affecting appearance. Exposed edges shall be straight and smooth, without protrusions. There shall be no sharp edges or corners exposed in the finished work which could be judged potentially harmful to wearing apparel or to the user.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on bed panels when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.4. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with ANSI Z1.4. The unit of product shall be one bed panel set. All bed panels offered for delivery at one time shall be considered a lot for the purpose of inspection.

4.3.1 Sampling for examination. Guidance for inspection level and an Acceptable Quality Level (AQL) is provided in 6.3.

4.4 Examination. Each bed panel shall be examined for defects listed in table I. Each attribute within each classification of multiple defects shall constitute one defect.

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TABLE I. Classification of defects.

Classification	Defects	Requirements Paragraph
Critical:	None defined.	
Major:		
101	Panel not complete, components missing.	3.1
102	Overall dimensions or corner radius not as specified.	Figure 1
103	Particleboard panel not as specified.	3.5.1
104	Laminated sheets not as specified.	3.5.2
105	Clamps not as specified.	3.5.4
106	Label holder not provided, when specified.	3.5.5 and 3.6.1
107	Plastic surfacing cracked, split, chipped, peeled, scratched, dented, or contains ripples.	3.6.1
108	Molding edge and panel edge not within tolerance specified.	3.6.1
109	Poor fit of component parts.	3.7
110	Sharp edges or protrusions in plastic, wood, or metal surfaces that could cause injury to tear clothing.	3.7
Minor:		
201	Screw missing, cracked, or split or threads stripped.	3.5
202	Screw head broken off or evidence fastener is not of type or size specified.	3.5
203	Plastic surfacing not pattern, color or finish specified.	3.5.2
204	Plastic edge molding not pattern or color to match plastic surfacing as specified.	3.5.1
205	Joint in plastic edge molding in wrong location.	3.6.1
206	Panels not located on footboards and headboards as specified, when installation is required.	3.6.2
207	Instruction sheets not provided, when procured without installation.	3.6.2
208	Components not neatly attached, poor fit or mating part, or parts misaligned or uneven affecting appearance.	3.7

5. PACKAGING

5.1 Packaging requirements. The preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES

(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The bed panels covered by this specification are intended for use as headboard and footboard panels with standard bunk beds conforming to AA-B-2892. Two panels are required per bed. The panels clamp directly to the upper and lower cross tubes of bed head and foot.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval (see 3.2 and 4.2.1).
c. Pattern, color, and finish required for decorative laminated plastic sheet (see 3.5.2).
d. Color of PVC coating on clamps if color other than gray is required (see 3.5.4).
e. Specify label holders, when required (see 3.5.5).
f. Specify if panel manufacturer is required to install the panels on the bunk beds (see 3.6.2).

6.3 Sampling for examination. Recommended inspection level II of ANSI Z1.4, and the AQL is 2.5 defects per hundred units (DHU) for major defects and 4.0 DHU for minor defects (see 4.3.1).

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article (see 3.2 and 4.2.1).

6.5 Drilling instructions. The distance between the top and bottom cross tubes of bunk beds vary among beds procured from different manufacturers, and between old design military bunk beds and the newer designs. Predrilling of panels by manufacturers is not recommended unless the exact dimensions can be determined for all beds for which the panels are being procured.

6.6 Selection of panel installer. The selection of an installer of the bed panels at the least cost to the Government should be determined by weighing factors specific to existing conditions at the time of the contract. Installation of the panels may be accomplished by the panel manufacturer, station personnel or a local manufacturer.

6.7 Condition and location of bunk beds. When installation is required, specify condition and location of bunk beds so as bidder can submit bids accordingly. The following information should be provided:

a. Are beds located in sleeping rooms or a central storage area?

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b. Are beds assembled or knocked down (KD)?
c. If KD, are bunk beds packaged or unpacked. If packed, who is to unpack?
d. If beds are in sleeping rooms, will rooms be occupied during installation?
e. Will installation of panels be restricted to certain hours? days?

6.8 Supersession data. This specification replaces Military Specification MIL-P-28636B dated 15 February 1985.

6.9 Part or Identifying Number (PIN). The specification number forms a PIN for bed panels covered by this document (see 1.1). PINs for the bed panels are established as follows:

Federal Specification Number AAP2930

MILITARY INTERESTS:

Custodians:
Army - GL
Navy - YD1

Review Activities:
Army - MD
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

DOJ
VA

PREPARING ACTIVITY:
Navy-YD1

(Project 7105-0288)

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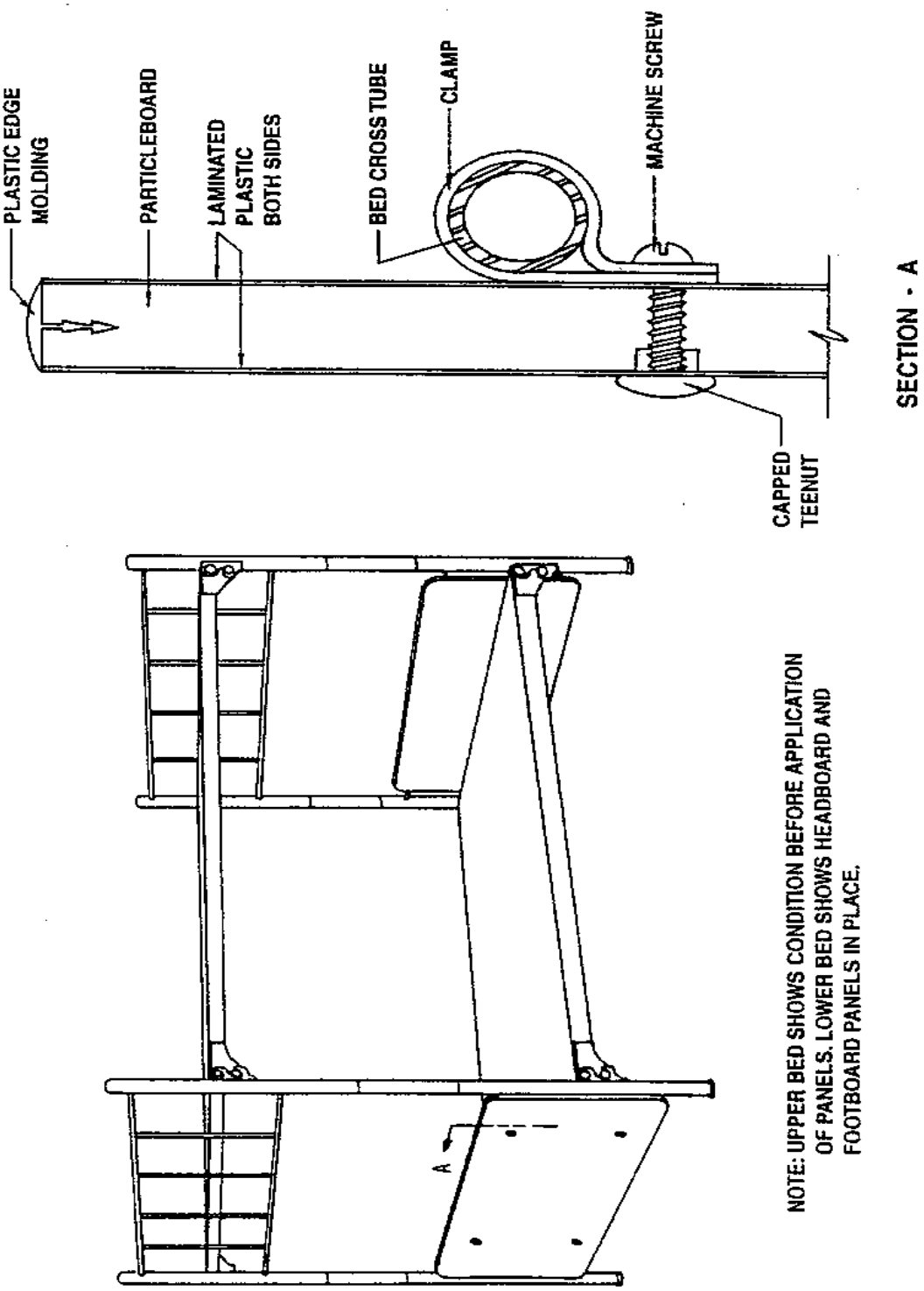


FIGURE 1. Headboard and footboard installation.

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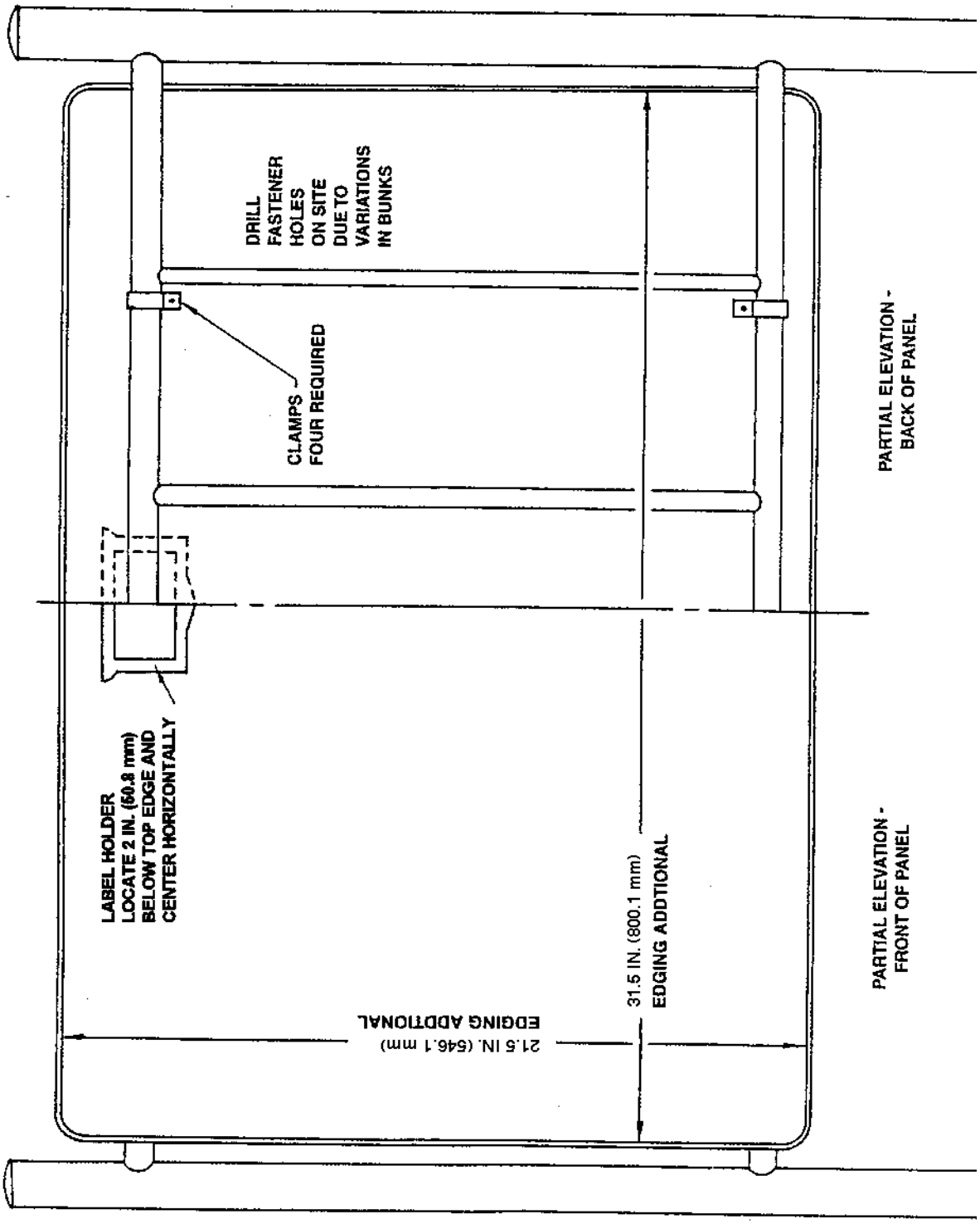


FIGURE 2. Bed panel installed.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	AA-P-2930	970715

3. DOCUMENT TITLE
PANEL FOR BUNK BED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *Include Area Code)*	
DANNY MUI	(1) Commercial 805-982-5666	(2) AUTOVON 551-5666
c. ADDRESS *(Include Zip Code)* COMMANDING OFFICER, NCBC CODE 15E2G 1000 23RD AVENUE PORT HUENEME, CA 93043-4301	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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